



Enterobius vermicularis (Pinworm)

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Why do pinworm symptoms peak at night, and what makes the gravid female migrate?
2. Which specimen and technique detect pinworm eggs — routine stool or something else?
3. How is mebendazole dosed for pinworm, and who should be treated?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Organism script — WHO / HOW

WHO	Children 5-14 years ; crowded settings (schools, pre-schools, orphanages, mental institutions, day-care); not tied to socioeconomic level . World's most common parasite; more common in females; ~1/3 are asymptomatic
HOW (transmission)	Hand-to-mouth — hands contaminated with embryonated eggs by scratching or by bedding, linens, night clothes; inhalation of airborne eggs in dust; rarely retro-infection through the anus
WHERE eggs pile up	Household dust — chairs, tables, pictures, floor; large numbers in the bedroom

2 Morphology — what to recognise

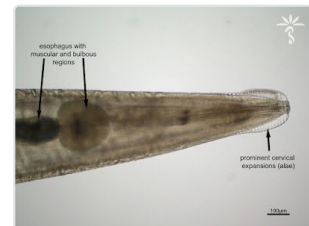
Feature	
Adult (both sexes)	Short, white, fusiform, thread-like ; mouth ringed by 3 transversely striated wing-like cervical alae ; double-bulb esophagus — unique to this worm
Female	8-13 mm ; pin-like tail (posterior third); gravid body filled with uteri holding thousands of eggs; oviparous; lives 5-12 weeks; lays 5,000-17,000 eggs
Male	2-5 mm ; tightly curved posterior end with a copulatory spicule; lives ~7-8 weeks
Egg	Planoconvex (flat one side, convex the other), 50-60 × 20-30 µm; colourless, not bile-stained ; thick double-layered transparent shell; floats in saturated salt ; coiled embryo becomes infectious ~6 h after deposition; viable ~2 weeks in cool moist conditions



Adult worms (thread-like).

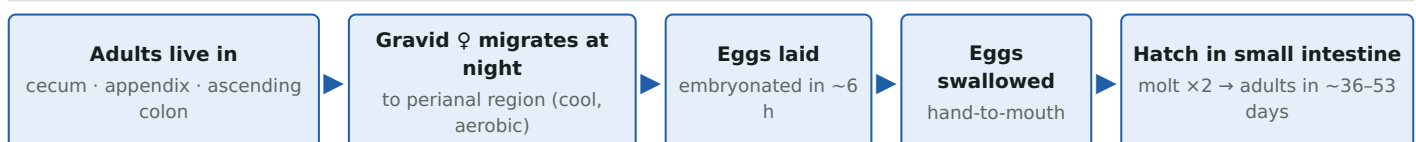


Eggs: planoconvex, thick shell.

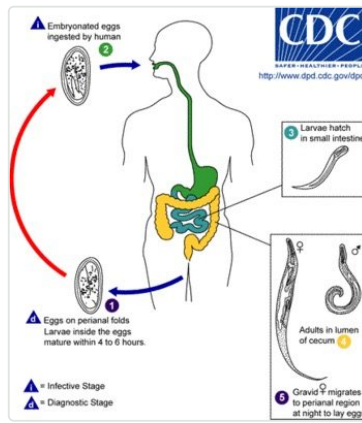


Cervical alae and bulbous esophagus.

3 Life cycle — the night-time cycle



Diagnostic stage = eggs on perianal folds; infective stage = embryonated egg.



Life cycle (CDC) — eggs on perianal folds mature within 4–6 hours.

4 Pathogenesis and clinical picture

Mechanism	Result
Female migrates + lays eggs at night	Perianal / perineal itching (pruritus ani) → scratching → bleeding and secondary bacterial infection
Nocturnal migration	Disturbed sleep; nocturnal enuresis sometimes
Adults attach to gut mucosa	Small ulcers, mild inflammation, bacterial infection (less common)
Worm enters vulva / vagina	Irritation and mucoïd discharge ; may reach uterus, tubes, peritoneum → chronic salpingitis, cervicitis, peritonitis, recurrent UTIs
Found in removed appendix	Claimed to be responsible for appendicitis
Blood / immune tests	No evidence of eosinophilia or raised serum IgE

5 Diagnosis · treatment · prevention

	Detail
Diagnosis	Characteristic eggs in the pre-anal area using adhesive cellophane tape ; in heavy infection the adult female can be seen at night in the pre-anal region
Treatment	Mebendazole 100 mg single dose, repeated two weeks later
Prevention	Cut nails short; wash hands after toilet and before meals; avoid scratching; ointment to reduce egg spread; clean contaminated surfaces; treat all household members

P Must-remember pearls

- Eggs are found by **adhesive cellophane tape** on the perianal skin — not by routine stool exam.
- Female migrates **at night** (cool, aerobic) → **pruritus ani**, poor sleep, sometimes enuresis.
- Egg = **planoconvex**, floats in salt; infectious **~6 h** after laying. Adult has **cervical alae + double-bulb esophagus**.
- Route: **hand-to-mouth** (also dust inhalation). **No eosinophilia / raised IgE**.
- **Mebendazole 100 mg, repeat at 2 weeks**; treat the whole household.

Q Test yourself

1. A 6-year-old has nocturnal perianal itching and bedwetting. Which test is most appropriate?

- A. Routine stool microscopy alone
- B. Adhesive cellophane tape applied to the perianal skin
- C. Blood eosinophil count
- D. Stool culture (Harada-Mori)

2. Microscopy shows an elongated ovoid egg, flat on one side and convex on the other, 50-60 × 20-30 μm. This is:

A. *Ancylostoma duodenale*
C. *Enterobius vermicularis*

B. *Taenia solium*
D. *Necator americanus*

3. A child is treated with mebendazole 100 mg once. What else does the lecture advise?

A. No further dose is needed
C. Give praziquantel next

B. Repeat the dose two weeks later and treat all household members
D. Treat only the child

4. A child scratches the perianal area then eats with unwashed hands. This illustrates which route?

A. Skin penetration by larvae
C. Ingestion of undercooked pork

B. Hand-to-mouth transmission of embryonated eggs
D. Retro-infection through the anus

5. Which finding is NOT expected in pinworm infection?

A. Perianal itching at night
C. Mucoid vaginal discharge in girls

B. Eosinophilia with raised serum IgE
D. Nocturnal enuresis

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B — Eggs are detected in the pre-anal area with adhesive cellophane tape.

2 — C — Planoconvex eggs are characteristic of *E. vermicularis*.

3 — B — Mebendazole is repeated after 2 weeks and household mass treatment prevents reinfection.

4 — B — Hands contaminated by scratching or bedding transmit embryonated eggs to the mouth.

5 — B — The lecture states there is no evidence linking enterobiasis to eosinophilia or raised IgE.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam